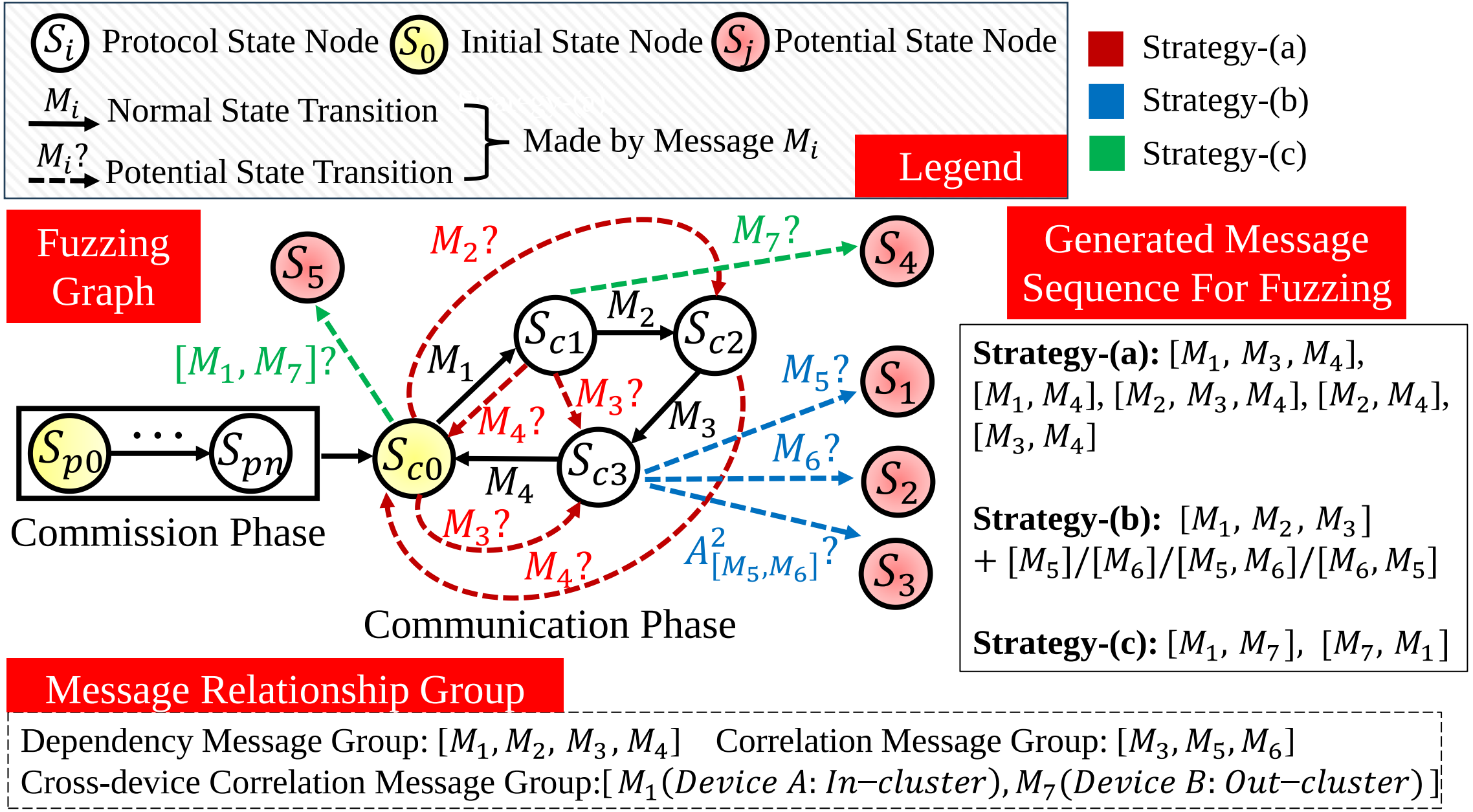

S_i Protocol State Node
S_0 Initial State Node
S_j Potential State Node
M_i Normal State Transition
$M_i?$ Potential State Transition
Made by Message M_i
Legend
Strategy-(a)
Strategy-(b)
Strategy-(c)
Fuzzing Graph
S_5
$M_2?$
$M_7?$
S_4
S_{c1}
M_2
S_{c2}
$[M_1, M_7]?$
M_1
$M_5?$
S_1
$M_4?$
$M_3?$
M_3
S_{p0} ... S_{pn}
S_{c0}
S_{c3}
$M_6?$
S_2
M_4
$M_3?$
$A^2_{[M_5,M_6]}?$
$M_4?$
S_3
Commission Phase
Communication Phase
Generated Message Sequence For Fuzzing
Strategy-(a): $[M_1, M_3, M_4]$, $[M_1, M_4]$, $[M_2, M_3, M_4]$, $[M_2, M_4]$, $[M_3, M_4]$
Strategy-(b): $[M_1, M_2, M_3]$ + $[M_5]/[M_6]/[M_5, M_6]/[M_6, M_5]$
Strategy-(c): $[M_1, M_7]$, $[M_7, M_1]$
Message Relationship Group
Dependency Message Group: $[M_1, M_2, M_3, M_4]$ Correlation Message Group: $[M_3, M_5, M_6]$
Cross-device Correlation Message Group:[$M_1(Device\ A: In-cluster), M_7(Device\ B: Out-cluster)$]